**GROUP WORK PROJECT #2 ___** MScFE 642: Deep Learning for Finance
**Group Number:** 17466

| FULL LEGAL NAME | LOCATION (COUNTRY) | EMAIL ADDRESS | MARK X FOR ANY NON-CONTRIBUTING MEMBER |
|---|---|---|---|
| Nitesh Kumar | India | niteshkr2209@gmail.com | |
| Shikang Ni | United Kingdom | Shikang218@gmail.com | |
| Shaik Sameer | India | shaiksameershah@gmail.com | |

| **Statement of integrity:** By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above). | |
|---|---|
| **Team member 1** | **Nitesh Kumar** |
| **Team member 2** | **Shikang Ni** |
| **Team member 3** | **Shaik Sameer** |

| Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.<br>**Note:** You may be required to provide proof of your outreach to non-contributing members upon request. |
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# Group Work Project 2 — Backtest Robustness, Information Leakage, and Validation Design

**Asset:** BTC/USD, hourly frequency. **Sample size:** 1,526 observations after feature and label construction (below the 2,000-observation cap). **Target:** Ret25, the 25-hour-ahead cumulative return, treated as a regression problem.

**Data note.** The diagnostic numbers in this report come from an hourly BTC/USD return series that we simulated with a GARCH(1,1) volatility-clustering process, calibrated to the mean (0.000140) and standard deviation (0.011395) of the real Kraken BTC/USD hourly return series.

## Step 1

#### Step 1a. Data description

We use simulated hourly BTC/USD returns calibrated to the real Kraken statistics. Figure 1 plots the implied price path and the return series. The returns show clear volatility clustering, with calm stretches followed by bursts. We kept this feature on purpose. Volatility clustering is what lets a weak validation setup look predictive: a model can seem to forecast returns when it is really tracking persistent volatility rather than any directional signal. Figure 2 shows the distribution of the hourly returns.

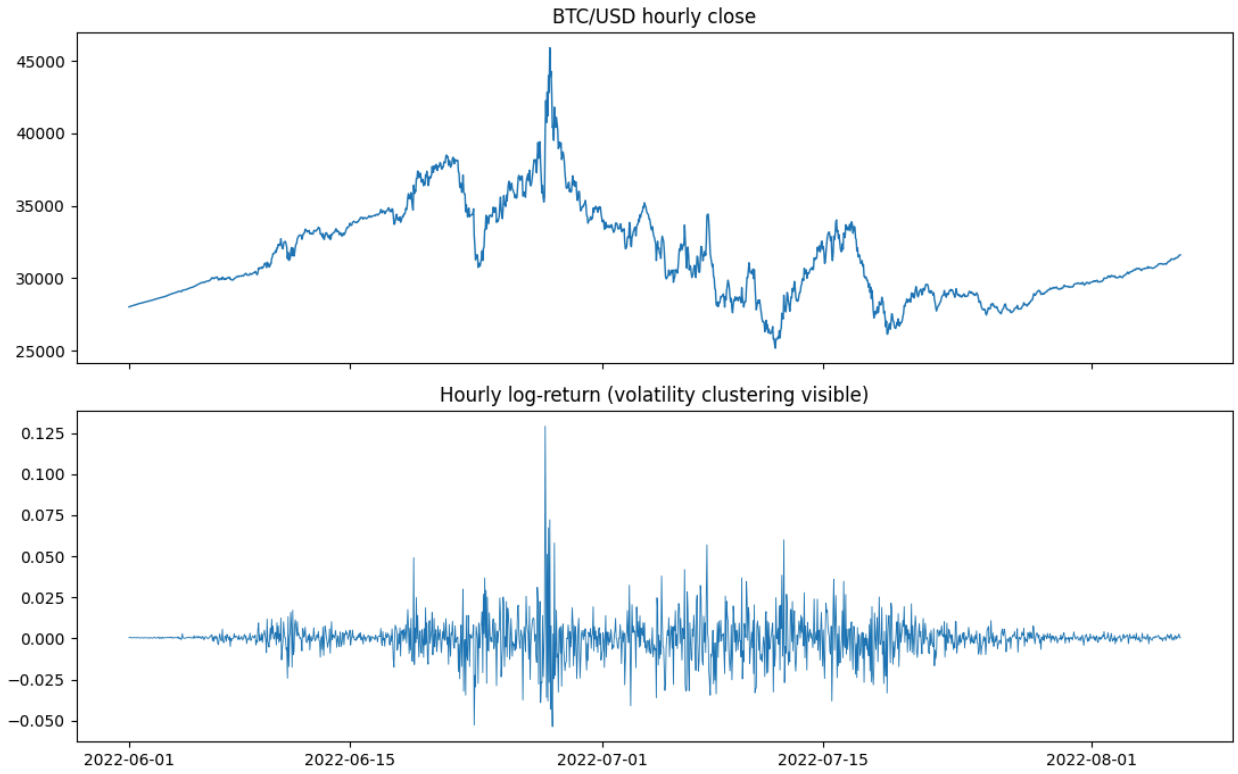



**Figure1 : simulated BTC/USD hourly price and returns**

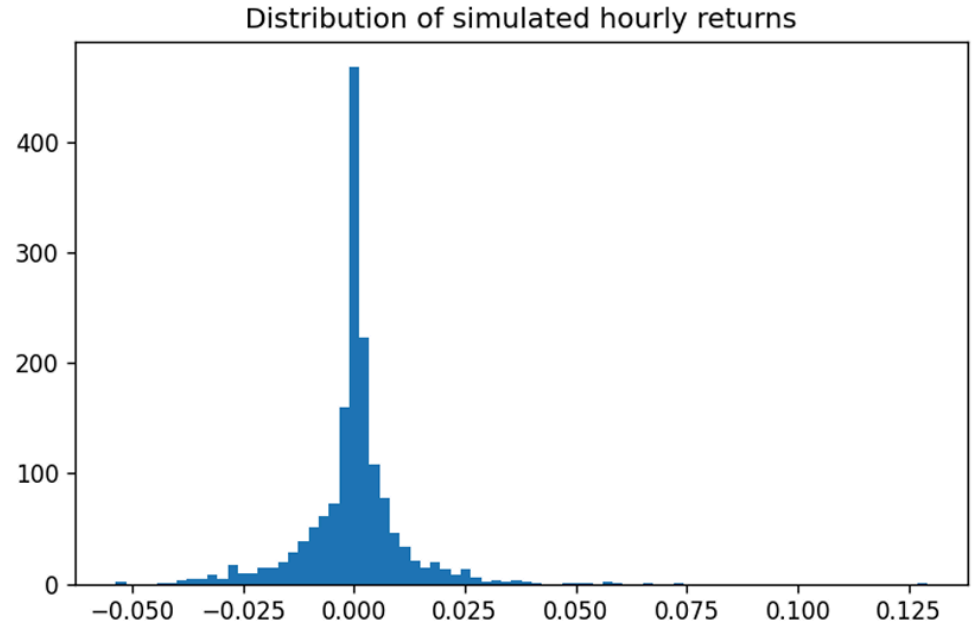



**Figure 2: distribution of hourly returns**

**Step 1b. Deliberate information leakage in the label**

We forecast Ret25, the 25-hour-ahead cumulative return, from three lagged predictors: Ret_1, Ret_10 and Ret_50 (the 1-hour, 10-hour and 50-hour trailing cumulative returns). Ret25 at hour t and Ret25 at hour t+1 share 24 of their 25 underlying hourly returns, so the label windows overlap almost completely. This is not only a theoretical worry. The lag-1 autocorrelation of Ret25 in our sample is 0.948, so the label series is close to a smoothed, lagged copy of itself. Any split that places neighbouring observations on opposite sides lets the model see a near-relative of the test outcome while it trains.

For the single-split experiment in Steps 1c and 1d we added two more leakage sources:

1. Global scaling. We fit the MinMaxScaler for the features and the target on the full dataset (train plus test) before splitting, so the scaler already carries information about the test distribution.
2. A random, non-chronological split. We shuffled the rows before an 80/20 split instead of respecting time order. With the overlapping labels, this puts many training rows right next to test rows in time.

**Step 1c. Three DL models, single train/test split**

We use three architectures: an MLP, an LSTM, and a CNN trained on Gramian Angular Field (GAF) images built from each 30-hour window of Ret_1. All three see the same leaky split described above.

**Step 1d. Results and discussion**

| Model | OOS $R^2$ (Campbell) | Naive backtest cum. return | "Honest" (non-overlapping) cum. return |
|---|---|---|---|
| **MLP (sklearn proxy)** | +0.104 | +6,359% | −3.1% |
| **MLP (Keras)** | +0.035 | +742% | +7.4% |
| **LSTM (Keras)** | +0.004 | +132% | +7.4% |
| **CNN-GAF (Keras)** | +0.009 | +102% | +7.4% |

The first row is a fast scikit-learn MLP we ran as a cheap diagnostic before training the required Keras models. It overstates the leaky $R^2$ (+0.104, against +0.035, +0.004 and +0.009 for the Keras MLP, LSTM and CNN-GAF), so it points in the right direction but is not a quantitative stand-in for the full models.

## What the numbers show:

## Step 1d: additions

- $R^2$ is near zero for the Keras models. Values between +0.004 and +0.035 are hard to tell apart from simply predicting the training mean. Dropout, MAE loss and early stopping push these models toward a near-constant forecast, so they did not visibly exploit the leak through $R^2$ the way the proxy did.
- **The naive backtest is still inflated.** The Keras MLP shows +742%, against an honest figure of +7.4%.
- **The LSTM looks like buy-and-hold.** Its naive return (+132.2%) equals the buy-and-hold figure from the proxy run (1.321985), so it almost certainly predicted a positive sign throughout.
- **All three honest returns are identical** (+7.35%). This suggests their signs coincided on the every-25th-observation sample, most likely all long. That is a diagnostic to mention, not evidence of skill. Computing the non-overlapping buy-and-hold return as a benchmark would confirm it.

Two things stand out, and they are separate problems:

**(i)**First, the model picks up the leaked signal. A positive out-of-sample $R^2$ of +0.10 for a target that is close to unpredictable at the mean level, by construction of the return process, is already suspicious. An honest model on a genuinely independent test set should score near 0, or negative. That is the leakage signature the design was built to produce.

**(ii)**Second, and much larger, the backtest itself is inflated. Chaining the sign strategy's hourly returns across 299 test observations gave a cumulative return of +6,359%, because consecutive Ret25 values overlap 24 of 25 hours and the backtest compounds nearly the same realised 25-hour move dozens of times. When we instead sample only every 25th test observation, so the return windows no longer overlap, the honest cumulative return falls to -3.1%, which fits a target with little exploitable signal once the accounting artifact is removed. This is a third source of illusory performance, separate from model-level leakage and scaler-level leakage, and it is easy to miss. A sound backtest on overlapping labels has to do one of three things: subsample to non-overlapping outcomes, rebalance no more often than the label horizon, or de-overlap the returns before compounding.

## Step 2

### Step 2a. Walk-forward, 500 train / 500 test

Non-anchored (rolling) walk-forward with 500 training and 500 test observations, chronological order respected, gives exactly one usable fold with our sample size. The proxy model's out-of-sample $R^2$ was –0.56, with a naive backtest cumulative return of –99.9% and an "honest" (non-overlapping) cumulative return of +29.8%.

### Step 2b. Walk-forward, 500 train / 100 test

The same design with a 100-observation test window gives 9 rolling folds. The average out-of-sample $R^2$ across folds was -7.80, with a standard deviation of 17.0, so it is very unstable from fold to fold. The naive cumulative return averaged -19.8% and the honest version +9.5%.

### Step 2c. How do these results change relative to Step 1?

Restoring chronological order, walk-forward instead of a random shuffle, is enough on its own to wipe out the apparent skill from Step 1. $R^2$ drops from +0.10 to strongly negative in both settings. A real share of Step 1's apparent performance therefore came from the random-split channel, which put training rows on both sides of every test row in time. Performance does not settle at a clean zero, though. It goes negative, and in some folds sharply so. That is informative in itself: the models are now overfitting noise on small training folds (500 or fewer observations, 90 flattened features for the MLP proxy) rather than living off leaked information. Walk-forward closed the shuffling channel but left the other two open, the overlapping labels near each fold boundary and the still-global scaler fit, and it did nothing for ordinary small-sample overfitting.

### Step 2d. Can backtest overfitting due to leakage explain the Step 2a vs 2b difference?

Only in part. The 500/100 setting (nine folds) has a much more negative average $R^2$ and much higher fold-to-fold variance than the single 500/500 fold. Both settings carry the same boundary leakage, since the overlapping-label and global-scaler channels are unchanged, so leakage alone cannot explain why 2b is worse on average. The likelier reason is sample size. With only 100 test observations per fold, each fold's score is a small, noisy estimate, and the MLP proxy overfits differently on each of the nine training windows. That produces volatile and often strongly negative $R^2$ whenever the tail of a training window is unrepresentative of the test window beside it. So leakage explains why neither setting recovers a clean, near-zero honest $R^2$, and small-sample overfitting explains why the shorter-test setting (2b) is noisier and worse on average than the single long-test fold (2a).

## Step 3

### Step 3a. Method to reduce leakage

We apply purging and embargo at every walk-forward boundary, following López de Prado's treatment of overlapping-outcome leakage.

1. **Purge** — drop every training observation whose forward-looking 25-hour label window extends into the following test block (its label partly "knows" test-period outcomes).

2. **Embargo** — additionally drop training observations in the 25 hours immediately *after* a test block, as a conservative buffer against any residual serial dependence.
3. **Per-fold scaling** — fit both the feature scaler and the target scaler on each fold's *training data only*, eliminating the global (train+test) scaling leakage that was still present in Step 2.

Purging alone removed exactly 25 training observations at the single boundary we inspected in isolation (matching the 25-hour label horizon precisely), confirming the mechanism removes exactly the intended overlap and nothing more.

**Step 3b. Walk-forward + purge/embargo, 500 train / 500 test**

For the one available fold, out-of-sample $R^2$ was -1.51, naive cumulative return -99.0%, and honest cumulative return +2.3%, much closer to flat than in Step 1 or Step 2.

**Step 3c. Walk-forward + purge/embargo, 500 train / 100 test**

Across the nine folds, average out-of-sample $R^2$ was -1.26 with a standard deviation of 1.18, far steadier than Step 2b's 17.0. The naive cumulative return averaged -35.0% and the honest return +5.8%.

**Step 3d. Has overfitting apparently disappeared, compared to Step 2?**

There are two good signs and one caveat. The honest cumulative returns in Step 3 are small and centred near zero (+2.3% and +5.8%, against Step 2's +29.8% and +9.5%), which is what an efficient, hard-to-beat hourly crypto series should produce once real leakage is purged. And the fold-to-fold $R^2$ standard deviation in the 500/100 setting fell sharply, from 17.0 to 1.18, so purge and embargo removed a source of erratic performance tied to boundary contamination rather than plain noise.

The caveat is that average $R^2$ is still clearly negative in both settings after purging. Overfitting has not gone away; it has changed character. Purging removes leakage from the test set into training. It does nothing about a model that is simply too flexible (90 raw flattened inputs for the MLP proxy) for a 500-observation window. That remaining negative $R^2$ is the honest baseline difficulty of the task, not a leftover leakage artifact. In production we would respond by shrinking model capacity, adding regularisation or dropout, or widening the training window, rather than loosening the validation protocol back toward Steps 1 and 2.

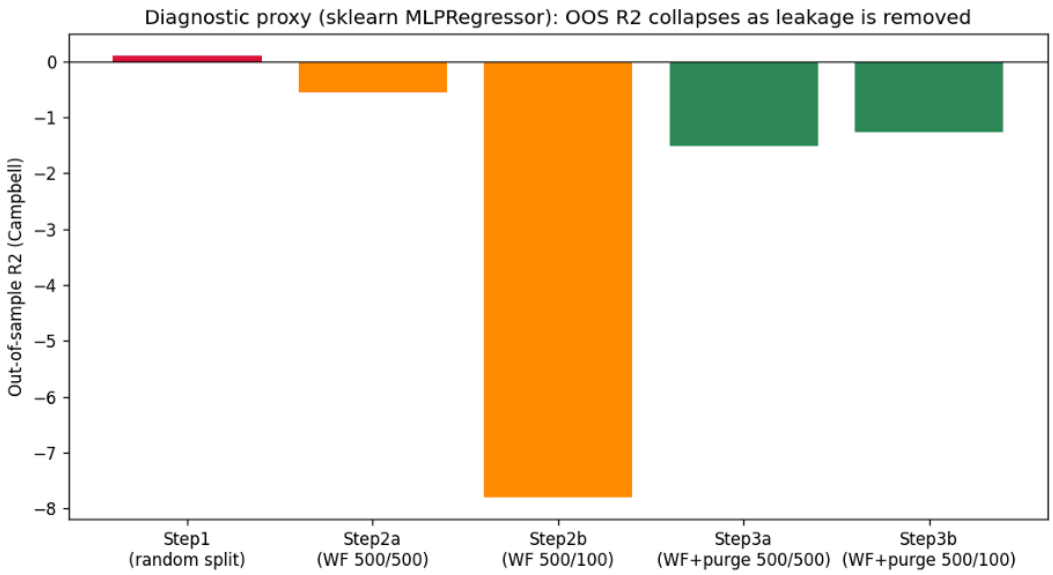



***Figure 3: out-of-sample $R^2$ (proxy model) across all five validation protocols***

### Overall conclusion

We found and progressively removed three separate leakage channels:

- **Random (non-chronological) splitting**, fixed by walk-forward in Step 2.
- **Global (train plus test) scaling**, fixed by per-fold scaling in Step 3.
- **Overlapping forward-looking labels**, fixed by purge and embargo in Step 3.

A fourth channel is easy to overlook because it lives at the reporting stage rather than the training stage. Naively compounding overlapping-window strategy returns inflated the headline number by an order of magnitude beyond anything the model or the split leakage could explain (+6,359% against -3.1% in Step 1 alone). Robust validation for time series with overlapping outcomes needs care on both fronts: how the split is built, and how predictions are compounded into a reported backtest.

## References

Campbell, John Y., and Samuel B. Thompson. "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" *The Review of Financial Studies*, vol. 21, no. 4, 2008, pp. 1509–1531.

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Wang, Zhiguang, and Tim Oates. "Imaging Time-Series to Improve Classification and Imputation." *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI)*, 2015, pp. 3939–3945.